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Section A – MCQs

1. (b) 8
2. (b) Contains all elements under consideration
3. (a) 4/5
4. (c) Many-one relation
5. (b) ½
6. (c) 60
7. (b) $A \cap B = \phi$
8. (c) Transitive

Section B – Very Short Answers

9. A set is a well-defined collection of objects.
Example: $A = \{1, 2, 3\}$.

10. Domain of $f(x) = x^2$ is all real numbers and range is $[0, \infty)$.

11. $\sin 45° \cos 45° = (1/\sqrt{2})(1/\sqrt{2}) = 1/2$.

12. $5P2 = 5 \times 4 = 20$.

13. Number of elements in power set of $A = 2^4 = 16$.

14. One-one function: $f(x)=x$, Many-one function: $f(x)=x^2$.

Section C – Short Answers

15. LHS $= \sin /(1+\cos\theta) + (1+\cos\theta)/\sin\theta$.
Taking LCM and simplifying, we get RHS $= 2 \operatorname{cosec}\theta$.

16. $A \cup B = \{1,2,3,4,5,6\}$ and $A \cap B = \{3,4\}$.

17. Total 3-digit numbers $= 5P3 = 5\times4\times3 = 60$.

18. Relation is reflexive, symmetric and transitive.

Section D – Long Answers

19. $(1 + \tan^2\theta)(1 + \sin^2\theta)$

$= \sec^2\theta(1 + \sin^2\theta)$

Using identity $1 + \tan^2\theta = \sec^2\theta$.

Hence LHS = RHS.